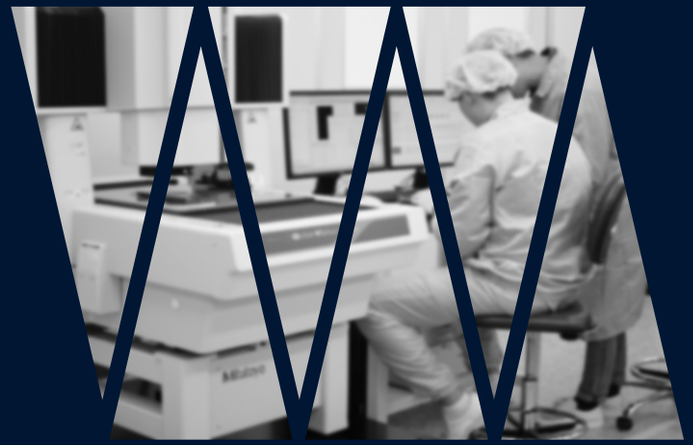
플라텍 주식회사

Platec Applied Material Co., Ltd.



저희 플라텍㈜ 는 약 30여년의 반도체 Engineering 경험을 바탕으로 열전도 Gasket , GND _ Shield (EMI) Gasket , 고기능성 Plastic , 다양한 수지류 부품 제조 공급하고 있습니다. 자사 임직원은 다 함께, 행복하게, 비전을 공유하는 회사의 가치를 실현하기 위해 노력하고 있으며, 고객사에는 최고의 제품 관리를 통해 원가 절감 과 품질 향상에 도움이 될 수 있는 협력사가 될 수 있도록 혼신의 노력을 다 하고 있습니다.

We Aid Technology | Platec Applied Material Co., Ltd.

경기도 용인시 기흥 구 흥덕 중앙로 120 #2201 Tel : 031-266-2445 E-mail : platec@platecamc.com

회사 소개

Information

- 회 사 명 : 플라텍㈜
- 대표이사 : 조 영 재
- 설립일자 : 2012년 7월 3일
- 주 소 :
  경기도 용인시 기흥 구 흥덕 중앙로 120 #2201
- 사업형태 : 반도체 특수 소재 개발/판매 및 Special Gasket 제조, 고기능성 수지 류 가공

PLALEZ ® O-ring 제품 소개

PLALEZ ® 제품 소개 자료

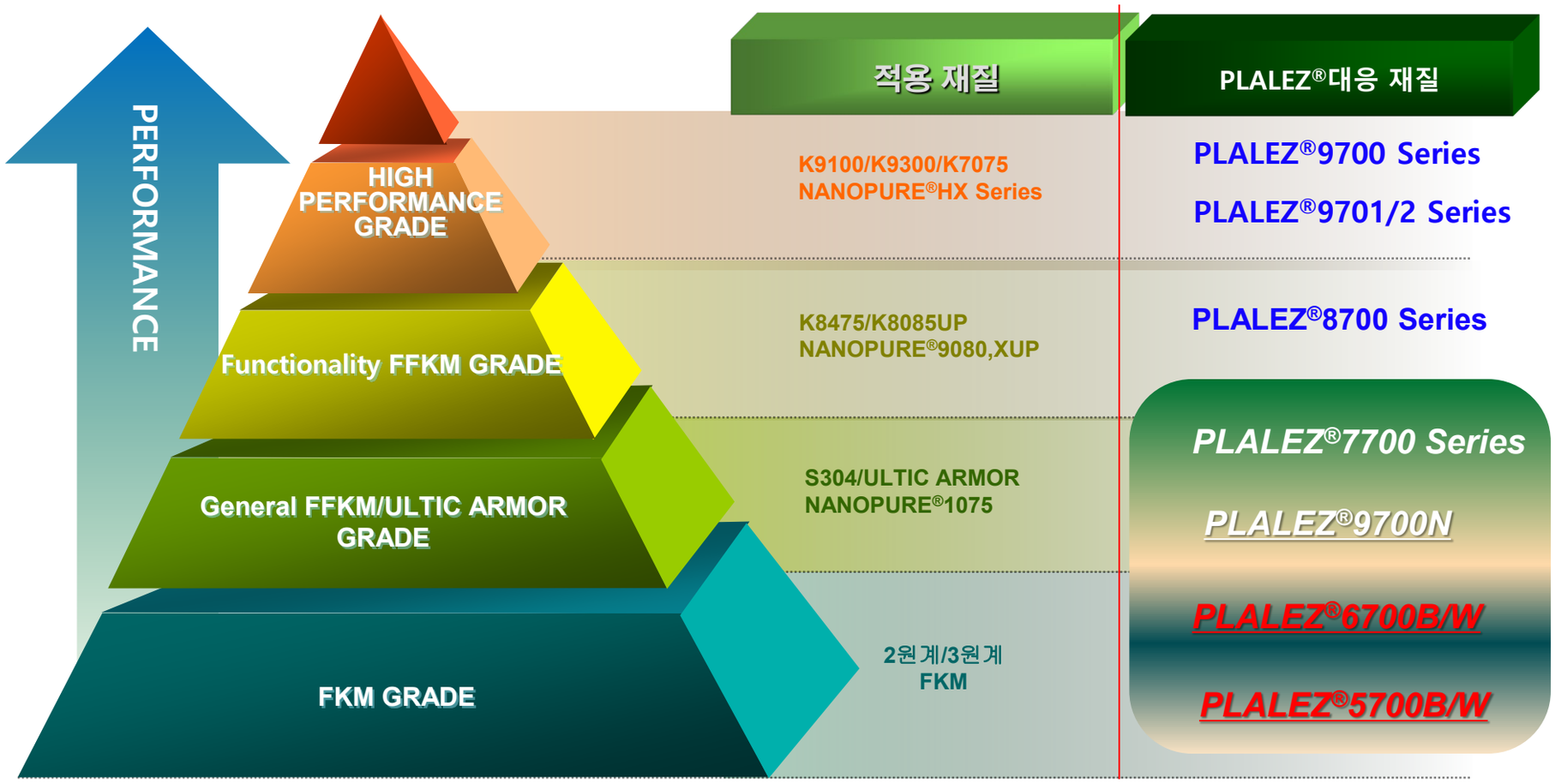
- PLATEC 제품 군
- PLALEZ ® Compounds Series
- PLALEZ ® Compounds 추천 Guide
- PLALEZ ® 제품 Catalog
  - PLALEZ ® 9700
  - PLALEZ ® 9702
  - PLALEZ ® 9700N
  - PLALEZ ® 8702
  - PLALEZ ® 7702

PLATEC 제품 군

PLALEZ® O-ring

PLALEZ ® Compounds Series

PLALEZ® O-ring



PLALEZ ® Compounds 추천 Guide

PLALEZ® O-ring

Contents	Process	Temp.	DuPont™ Kalrez®	GreeneTweed™ Chemraz®	PLATEC™ PLALEZ®	Feature
Plasma Processes	HDP-CVD / PE-CVD	25~200℃	9100 8085	XRZ 657	9700 9702 9700N	• **9700** 고온 공정에 탁월한 성능 Particle 억제 • **9701** 높은 내열성을 요구하는 공정에 적합 • **9702** 고온, Plasma 및 Gas 증착 공정 등에 사용 • **9700N** Plasma 공정 등에서 낮은 particle을 요구하는 구동부위에 적합 • **8702** 강한 Plasma 내성 및 Dynamic seal 수명 향상
	Etching	25~200℃				
	Ashing / Stripping	25~250℃	9100 8085	XRZ 657	9700 8702	
Thermal Processes	SA-CVD	25~250℃	8085	E38	9702 8702	
	Metal-CVD ALD, LP-CVD	25~300℃	8900 9100	657	9700 9701	
	Oxidation, Diffusion	150~300℃	8900 8475	657	9701 8702	
	Lamp Anneal, RTP	150~300℃	8475	E38	8702	

PLALEZ ® Compounds 추천 Guide

PLALEZ® O-ring

Contents	Process	Temp.	DuPont™ Kalrez®	GreeneTweed™ Chemraz®	PLATEC™ PLALEZ®	Feature
Wet Processes	Wafer Prep,	25~125℃	6375 4079		7701 7702	• **7701/7702** 흰색/검정의 재질로 복사열에 유리 모든 습식 화학 공정에 사용
	Etching	25~180℃				
	Photo lithography	25~125℃				
	Stripping	25~125℃	6375			
	Copper Plating	25~100℃				

PLALEZ ® 9700

PLALEZ ® 9700

For Semiconductor PECVD and HDPCVD Processes

PLALEZ® 9700는 반도체 식각(Etching) 및 증착(Deposition) 공정을 위해 특별히 개발되어진 초 순수 과 불소 제품입니다.
PLALEZ® 9700는 특히 가혹한 플라즈마 환경에서 낮은 식각율로 파티클 발생으로 인한 문제를 최소화하기 위하여 설계되었습니다. 또한 우수한 열적 안정성, 매우 낮은 outgassing뿐만 아니라 뛰어난 영구압축복원율 및 기계적 특성을 제공합니다.
PLALEZ® 9700S는 최대 325℃에서 사용 가능합니다.

Features & Benefit

- Low erosion rate and ultra-low particle generation in oxygen and fluorine-based plasmas
- Excellent reistance to dry process chemistry
- Excellent thermal stability
- Very low outgassing properties and metals content
- Excellent elastic recovery and low compression set properties

Suggested Applications

- Gas inlet/orifice seals
- Chamber lid / Isolation valve seals

Typical Physical Properties

Color	Dark Auburn
Polymer Type	Perfluoroelastomer
Hardness[1], Shore A	74
Tensile Strength at Break[2], MPa	12.4
Elongation at Break[3], %	178
100% Modulus[4], MPa	5.2
Compression Set[5], %	
70 hours @ 204℃	8
70 hours @ 250℃	12
70 hours @ 300℃	19
70 hours @ 320℃	34
Maximum Continuous Service, Temperature, ℃	320

1) ASTM D1415 (ISO 48)
2) ASTM D412 (ISO 37)
3) ASTM D412 (ISO 37)
4) ASTM D412 (ISO 37)
5) ASTM D395 Method B ; (ISO 815)

PLALEZ ® 9700

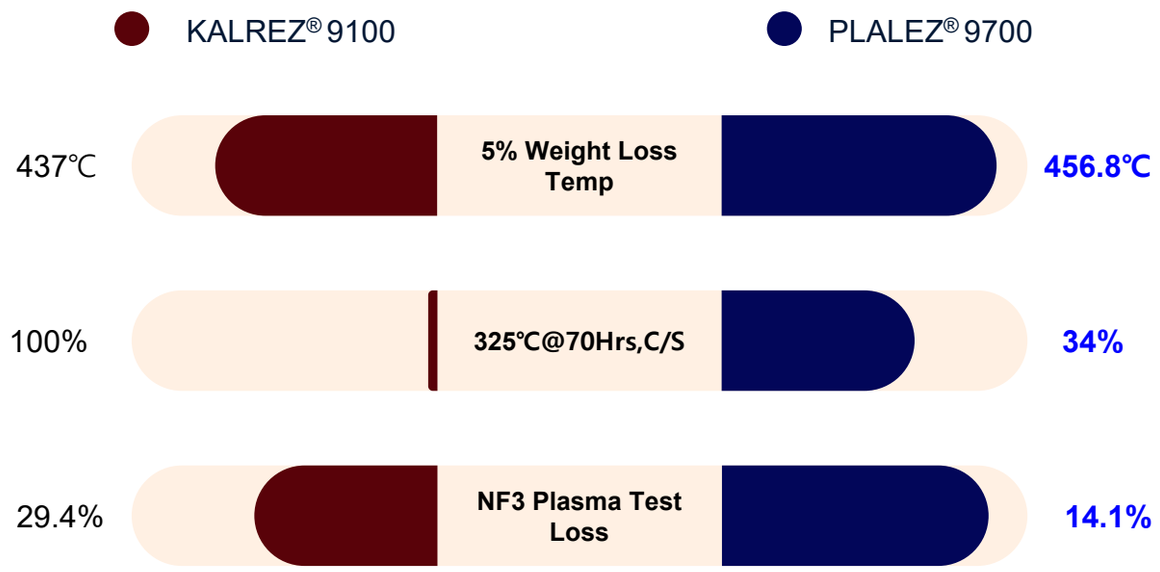
High Maximum Performance Compounds Series

PLALEZ ® 9700

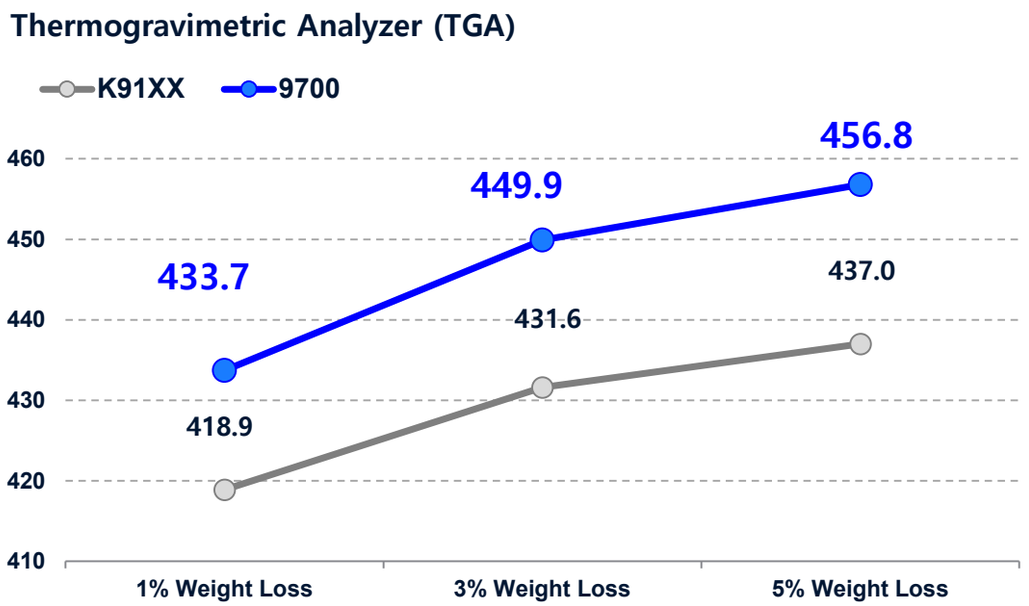
- High Performance, High Purity Grade
- Excellent high temperature stability (Max. 325℃)
- Excellent plasma resistance (Particle Zero)
- Expectation of cost savings compared to foreign production
- Life-Time Enhancement

PLALEZ ® 9700

✓ Material Properties _PLALEZ®9700



➔ Temperature characteristics are also better than KALREZ®9100



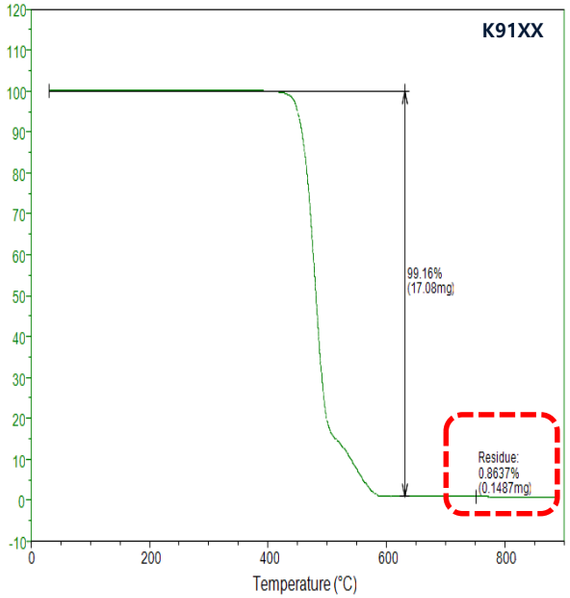
* Conditions : Equilibrate 30.00℃ under N2 → Ramp 10.00℃/min to 600.00℃ → Change the gas to Air → Ramp 10.00℃/min to 900.00℃

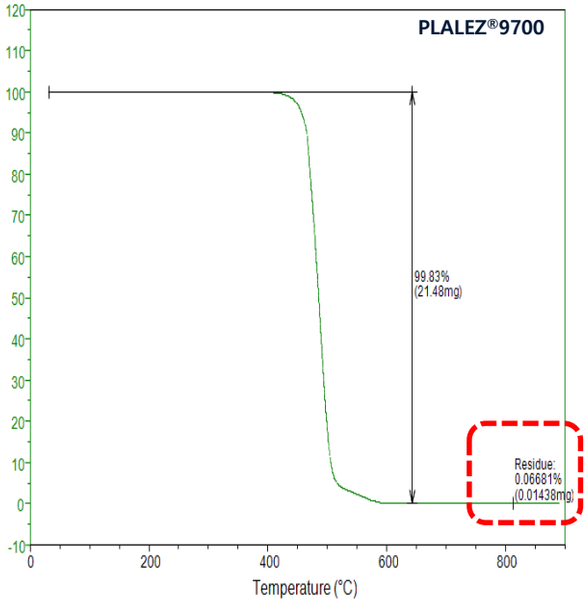
PLALEZ ® 9700

✓ Material Properties _PLALEZ®9700

- Very little inorganic residue

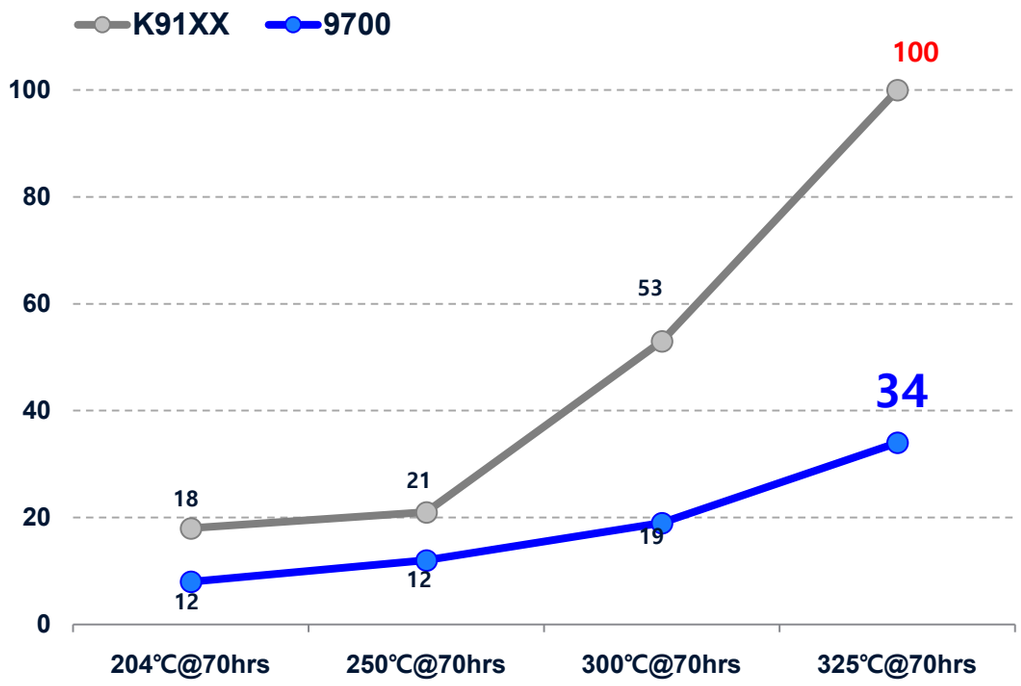
	KALREZ®K9100	PLALEZ®9700
Residual Occupancy (%)	0.8637 %	0.06681 %
Residual amount (mg)	0.1487 mg	0.01438 mg





* Conditions : Equilibrate 30.00℃ under N2 → Ramp 10.00℃/min to 600.00℃ → Change the gas to Air → Ramp 10.00℃/min to 900.00℃

- Compression Set Comparison

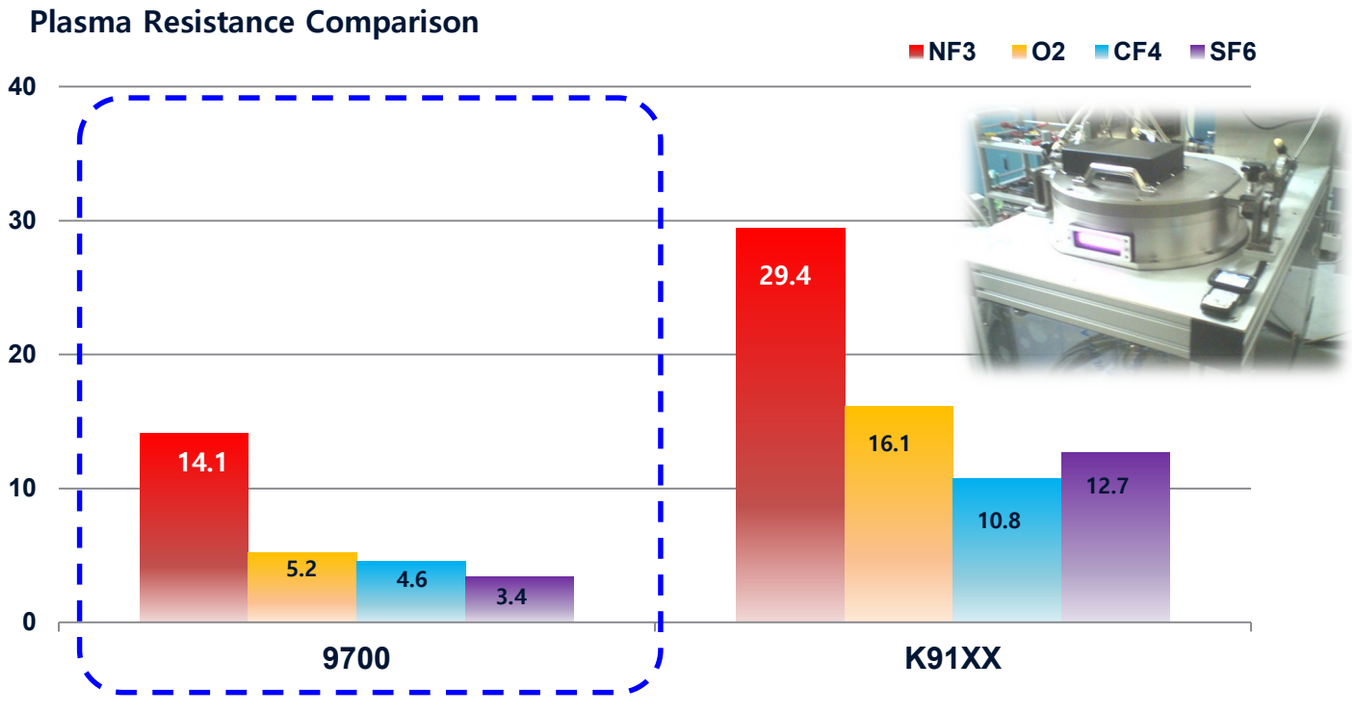


* Conditions : ASTM D395 Method B (AS568A K214 O-Ring test specimens)

PLALEZ ® 9700

✓ Material Properties _PLALEZ®9700

➔Plasma resistance characteristics in process gas are more than twice as good as k9100



* Conditions : RF power 800W, gas flow 100sccm, temperature 200℃, test time 30min

PLALEZ ® 9702

PLALEZ ® 9702

High Temperature resist FFKM Parts For High performance Sealing Compound For Semiconductor

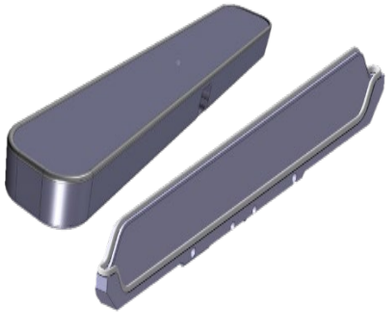
Pure perfluoro-elastomer developed with high-performance materials for use in harsh processes in the semiconductor and display industries

Features & Benefit

- Excellent resistance to "dry" gas process environments
- Very low weight loss in reactive plasmas
- Excellent (low) compression set properties
- Excellent elastic recovery properties

Suggested Applications

- Chamber lid seals
- Gas inlet seals
- Quartz window seals
- Other thermal applications
- Other plasma applications



Typical Physical Properties

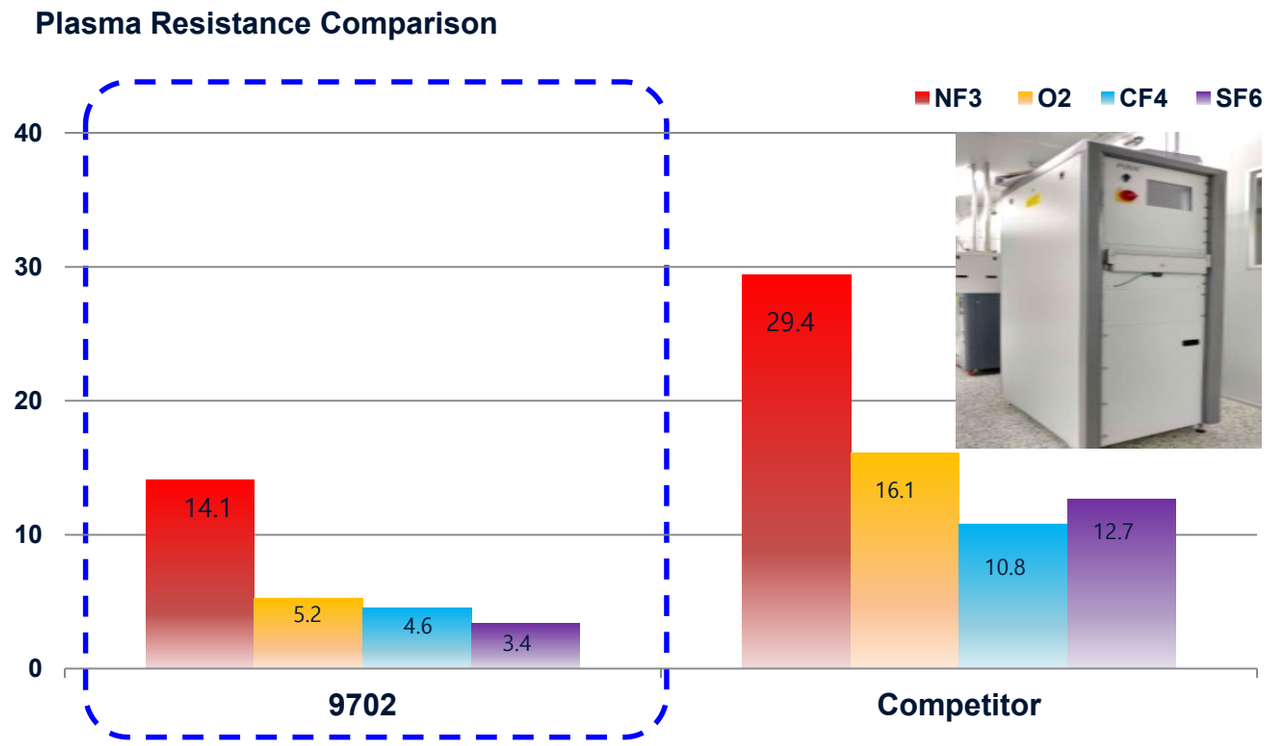
Service Temperature	-20℃ ~ 320℃
Polymer Type	FFKM
Color	Off-White
Physical Properties :	
Hardness[1] Shore A	75
Tensile Strength[2] (Mpa)	10.0
Elongation[2](%)	220
Modulus[2] @100%	4.6
Compression Set[3](%) at 204℃ 70 hrs	16
Compression Set[3](%) at 300℃ 70 hrs	32.6
Service Temperature(℃)	320

1. ASTM D2240(AS568 214 O-Ring test specimens)
2. ASTM D412(AS568 214 O-Ring test specimens)
3. ASTM D395B(AS568 214 O-Ring test specimens)

PLALEZ ® 9702

✓ Material Properties _PLALEZ®9702

➔Plasma resistance characteristics in process gas are more than twice as good as kxxx



* Conditions : RF power 800W, gas flow 100sccm, temperature 200℃, test time 30min

PLALEZ ® 9700N

PLALEZ ® 9700N

PLALEZ ® 9700N는 반도체 공정용으로 특별히 개발되어진 초 순수 제품입니다. PLALEZ ® 9700N는 특히 가혹한 플라즈마 환경에서 낮은 식각율로 파티클 발생으로 인한 문제를 최소화하기 위하여 설계되었습니다. CVD 및 건식 플라즈마 Etch 장비용으로 적합합니다.
PLALEZ ® 9700N 는 최대 250℃에서 사용 가능합니다.

Features & Benefit

- High purity (Particle free) very low metals content
- Excellent plasma resistance
- Excellent abrasion resistance
- Very low metals content
- Low compression set

Suggested Applications

- Dry etch equipment
- Plasma ashing equipment
- Bonded gate valves / slit valve door seals
- Gas inlet seals / Chamber lid seals

Typical Physical Properties

Color	Translucent
Polymer Type	FHPE
Hardness[1], Shore A	68
Tensile Strength at Break[2], MPa	20.0
Elongation at Break[3], %	280
100% Modulus[4], MPa	2.6
Compression Set[5], %	
70 hours @ 200℃	20
70 hours @ 250℃	37
Maximum Continuous Service, Temperature, ℃	250

1) ASTM D1415 (ISO 48)
2) ASTM D412 (ISO 37)
3) ASTM D412 (ISO 37)
4) ASTM D412 (ISO 37)
5) ASTM D395 Method B ; (ISO 815)

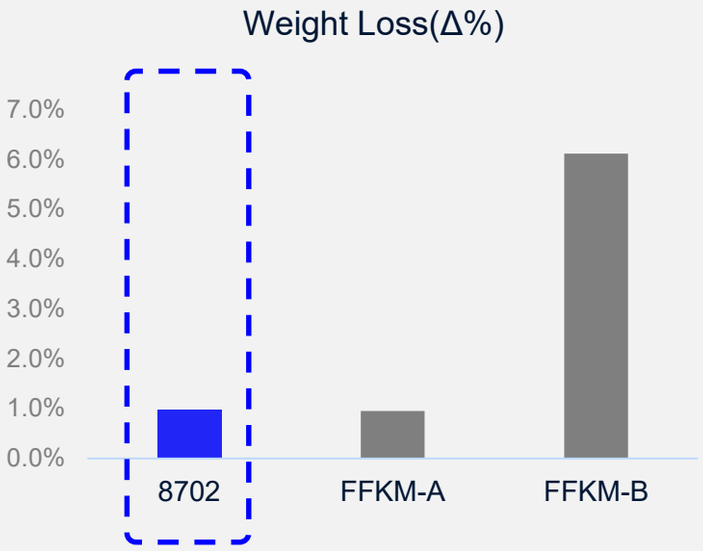
PLALEZ ® 8702

PLALEZ ® 8702

High Temperature resist FFKM Parts For Next Generation Sealing Compound For Semiconductor.

Excellent mechanical properties withstands a high temperature resistance.

PLASMA TEST RESULT



*** PTP-300 Plasma Asher test condition is used gas for O2 200sccm, CF4 100sccm Vacuum Pressure 1.5torr,RF Power 500watt, Temperature 200 ℃, Test time10 min/10cycle.**

Typical Physical Properties

Service Temperature	-20℃~290℃
Polymer Type	FFKM
Color	White
Physical Properties :	
Hardness[1] Shore A	72
Tensile Strength[2] (Mpa)	18.0
Elongation[2] (%)	167
Modulus[2] @100%	6
Compression Set[3](%) at 204℃ 72 hrs	17.7
Compression Set[3](%) at 250℃ 72 hrs	25.1
Service Temperature(℃)	290

1) ASTM D2240(AS568 214 O-Ring test specimens)
2) ASTM D412(AS568 214 O-Ring test specimens)
3) ASTM D395B(AS568 214 O-Ring test specimens)

PLALEZ ® 7702

PLALEZ ® 7702

PLLEZ®7702 is an upgraded FFKM new material that maximizes heat resistance while maintaining the basic characteristics of PLLEZ®775M

Minimal contamination withstands a variety of aggressive Chemicals at Elevated Temperatures. excellent refractory properties and is capable of providing a variety of harsh chemicals
Designed to withstand, and with excellent mechanical properties, both static and dynamic processes
Applicable, suitable for slit valves.

Features & Benefit

- Withstands a Variety of aggressive chemicals
- Outstanding Oil Resistance
- Outstanding Solvent Resistance
- Excellent Mechanical Properties
- Good Compression set Resistance

Suggested Applications

- Gas Inlet Seals
- Slit Valve Door Seals / Bonded Gate Valves
- Chemical Lid Seals

It has been developed specifically with a modified high-purity polymer to minimize contamination by upgrading the existing PLLEZ®775M, using the existing characteristic of minimal PTFE fillers and using the semiconductor's high-density O2, CF4 etching plasma.

Typical Physical Properties

Service Temperature	-20℃~270℃
Polymer Type	Perfluoro elastomer
Color	White
Hardness Shore A[1)]	75
Tensile Strength (Mpa) [2)]	16.0
Elongation(%)[2)]	200
Modulus@100%[2)]	8
Compression Set[3)](%) at 250℃ 72 hrs	28
Compression Set[3)](%) at 200℃ 72 hrs	18
Maximum Temperature, ℃	270

1) ASTM D2240(AS568 214O-Ring test specimens)
2) ASTM D412(AS568 214O-Ring test specimens)
3) ASTM D395B(AS568 214O-Ring test specimens)

감사합니다